MIL-D-11303D
26 August 1971
SUPERSEDING
MIL-D-11303C
12 August 1965

MILITARY SPECIFICATION

DRUM-HANDLING ATTACHMENT, FORK LIFT TRUCK,

(FOR HORIZONTAL HANDLING)

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a carriage mounted drum-handling attachment for use in lifting 55-gallon drums in the horizontal position (see 6.1).

1.2 Classification. Drum-handling attachment shall be of the following type:

Type I - Three drum capacity - carriage mounted - prong type.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-P-416	- Plating, Cadmium (Electrodeposited).
PPP-C-650	- Crates, Wood, Open and Covered.
PPP-D-729	- Drums: Metal, 55-Gallon (for Shipment of Noncorrosive Material).
PPP-T-97	- Tape, Pressure-Sensitive Adhesive, Filament Reinforced.

Military

MIL-P-116	- Preservation, Methods of.
MIL-P-514	- Plate, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.

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STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of U. S. Military Property.

DRAWING

ME

DL13216E9938	- Drum Handling Attachment, Fork Lift Truck: Type I.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28	- Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 202, Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC. AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street, N. W., Washington, D. C. 20036.)

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AMERICAN WELDING SOCIETY

Welding Handbook
Standard Qualification Procedure

(Application for copies should be addressed to the American Welding Society, 345 East 47th Street, New York, N. Y. 10017.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, N. Y. 10017.)

3. REQUIREMENTS

3.1 Description. The drum-handling attachment (hereinafter called attachment) shall be capable of handling filled 55-gallon drums in the horizontal position by means of a fork-lift truck. The type I attachment consists of a series of specially-shaped and spaced forks that cradle one, two, or three drums to be handled.

3.2 First article.

3.2.1 Preproduction model. The supplier shall furnish an attachment for examination and test within the time frame specified to prove, prior to starting production, that his production methods, will produce attachments that comply with the requirements of this specification. Examination and tests shall be those specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Materials. Material shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.4 Fastening devices. All fastening devices shall be fabricated of steel and shall be cadmium plated in accordance with QQ-P-416, type II, class 3. Threads shall conform to Handbook H28.

3.5 Type I.

3.5.1 Design and construction. The design and construction of all components and assemblies of the type I attachment shall conform to the requirements herein and the design, details, specified dimensions, tolerances and sizes as shown on DL13216E9938. The outer prongs shall be formed from steel bar stock as specified. The inner prongs shall be either drawn, forged, swaged, or spun to the shape shown on the drawing.

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3.5.2 Drawings. The drawings forming a part of this specification are engineering design drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the supplier or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.5.3 Performance. Attachments shall have a rated capacity of 1,950 pounds and shall be capable of sustaining 150 percent of this rated load. When tested as specified in 4.5.2.1 there shall be no evidence of the following:

(a) Fracture of welds.
(b) Permanent deformation of prongs.
(c) Bending, deformation, fracture, or binding of hanger strap.

3.6 Identification marking. The attachment shall be identified in accordance with MIL-STD-130. The marking shall be applied to the attachment on plates conforming to MIL-P-514, type I, style 3, composition C, of type I, grade A, class 1 material. Plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.7 Instructions. Unless otherwise specified, each attachment shall be furnished with installation instructions and an instruction sheet covering operation, replacement parts and maintenance.

3.8 Treatment and painting. The attachment shall be cleaned, treated, and painted in accordance with MIL-T-704, type A. Unless otherwise specified, the color of the finish coat shall be olive drab, color No. 24087.

3.9 Workmanship. The completely fabricated attachment shall conform to the quality and grade of product established by this specification, and shall be clean and correctly assembled. All bends shall be made with metal dies or fixtures. Burrs shall be removed and sharp edges shall be rounded. The attachment shall show no evidence of rust or corrosion.

3.9.1 Rivets. Rivets shall completely fill the rivet holes. Rivet heads shall be flush to the joining member and shall not be cracked, fractured, or broken.

3.9.2 Fastening devices. All nuts, bolts, pins and similar parts shall be installed with adequate means for preventing them from loosening or becoming out of adjustment. All fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

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3.9.3 Welders and welding.

3.9.3.1 Welders. Before assigning any welder to manual welding work covered by this specification, the supplier shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

Standard Qualification Procedure of the American Welding Society.
Welding Qualifications of the ASME.

3.9.3.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading.

3.9.4 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more parts possessing such functional and physical characteristics as to be equivalent in performance and durability and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts except for adjustment, and without selection for fit or performance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings.

4.1.2 Disassembly inspection. Failure of any test by the preproduction model shall be cause for disassembly, in the presence of a Government representative, of the preproduction model to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected.

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4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of preparation for delivery (see 4.6).

4.3 Preproduction inspection.

4.3.1 Examination. The attachment shall be examined as specified in table I. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The attachment shall be tested as specified in 4.5.2 and 4.5.2.1. Failure of any test shall be cause for performing the inspection specified in 4.1.2.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in table I. The AQL shall be 2.5 percent defective.

4.4.3 Tests.

4.4.3.1 Samples. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2 and 4.5.2.1. Failure of any test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. Examination shall be in accordance with table I.

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Table I. Classification of defects

	Examine	Defect
101	Finish	Missing.
102		Blistered, peeled, chipped, scratched, or foreign matter imbedded.
103		Indication of rust or corrosion.
104	Design (Type I)	Components do not conform to the design and details as shown on the drawing.
105	Construction and workmanship (applicable to components and assemblies)	Sharp burrs or edges.
106		Fractured, split, sprung, cracked, malformed, dented, bowed, or broken.
107	Construction and workmanship (applicable to components and assemblies)	Welding, brazing, or soldering not smooth, has deformation of metal or rough projecting edges.
108		Rivets, cracked, fractured, or broken.
109		Bolts, washers, or nuts not sealed properly.
110		Fastening device not secured.
111		Fastening devices subject to removal are swaged, peened, staked, or otherwise permanently deformed.
112		Forgings not free from scale, inclusion, cold shuts, or seams (when applicable).
113		Dimensions not as specified on the drawing.
114	Instructions	Missing, do not cover mounting, operation, and maintenance, not legible.
115	Marking for identification	Missing, incomplete, not legible.

4.5.2 Tests. Unless otherwise specified herein, all tests shall be performed using a 2,000 or 4,000 pound capacity at 24-inch load center, solid rubber tired fork lift truck, and containers conforming to PPP-D-729.

4.5.2.1 Type I. Each sample type I attachment shall be tested as follows to determine compliance with 3.5.3:

(a) Mount attachment to fork lift truck or simulated rigging device of suitable capacity.

(b) Pick up three filled 55-gallon drums and such additional equally distributed load as is necessary to insure the attachment is supporting at least 150 percent of rated load. Hold the load for five minutes and lower to the ground. Inability to mount the attachment securely or noncompliance with 3.5.3 shall constitute failure of this test.

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4.6 Inspection of preparation for delivery.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completely processed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

116.	Materials, methods, and containers not as specified for level A. Each incorrect material, method, or container shall constitute a defect.
117.	Unpainted exterior metal surfaces not coated with preservative as specified for level A.
118.	Securing not as specified for level A.
119.	Instruction sheet(s) not preserved and secured to the attachment as specified for level A.
120.	Blocking, bracing, and anchoring not in accordance with the referenced document for level A.
121.	Closure and strapping not in accordance with the referenced document for level A.
122.	Marking missing, illegible, incorrect, or incomplete.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified.

5.1.1 Level A.

5.1.1.1 Unprotected surfaces. Unpainted exterior metal surfaces of components requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P-1 preservative. The preservative shall conform to the applicable specification listed in and shall be applied in accordance with MIL-P-116.

5.1.1.2 Securing. Clamp screws and wing nuts shall be screwed tight. All moveable parts shall be secured in a fixed position in a manner to prevent movement while in transit.

5.1.1.3 Instruction sheet(s). The instruction sheet(s) shall be preserved in accordance with MIL-P-116, method IC-1 and secured to the attachment in a protected location with tape conforming to PPP-T-97, type I, size optional.

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5.1.2 Level C. Each complete attachment with instruction sheet shall be preserved and packaged to afford protection against deterioration and damage from the supplier to the initial destination.

5.2 Packing. Packing shall be level A or C as specified.

5.2.1 Level A. Each attachment, preserved and packaged as specified in 5.1, shall be packed in a close-fitting crate conforming to PPP-C-650, style B, type II. Blocking, bracing, and anchoring of the attachment and crate closure and strapping shall be in accordance with the appendix to the crate specification.

5.2.2 Level C. The attachments, preserved and packaged as specified in 5.1, shall be packed to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The attachment is intended for use with fork lift trucks to handle 55-gallon metal drums used in shipment and storage. The attachment is used to lift, transport, tier, and unload drums in a horizontal position.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of the preproduction model (see 3.2.1).
(c) Technical instructions if other than specified (see 3.7).
(d) Color of finish coat if other than as specified (see 3.8).
(e) Level of preservation and packaging, and level of packing required (see 5.1 and 5.2).

6.3 Preproduction model. Any changes or deviations of production attachments from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish attachments conforming to this specification.

6.4 Type II. Type II drum-handling attachments have been discontinued since they are no longer required.

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Custodians:

Army - ME
Navy - SA
Air Force - 84

User Activities:

Army - CE, SM
Navy - MC

Preparing Activity:

Army - ME

Project No. 3930-0118

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is *provided for obtaining information* on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-D-11303D - Drum-Handling Attachment, Fork, Lift Truck, (For Horizontal Handling)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 65 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

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